CENTRAL COLLEGE SCHOOL BULENGA NATETE CAMPUS

END OF TERM ONE 2026

UGANDA ADVANCED CERTIFICATE OF EDUCATION

P510/1 PHYSICS THEORY EXAMINATION PAPER

Time: 2 hours and 40 minutes

Instructions:

1. Attempt **four** questions in total. Each question carries **20 marks**.
2. You must attempt **one question from each of the four sections**:
3. Section A is compulsory.
4. Use the provided data and support materials to guide your solutions.
5. Clearly show all calculations and assumptions.

SECTION A

ITEM 1

At a regional referral hospital in Kampala, a newly installed x-ray imaging unit is being tested. the radiographer explains to trainees that x-rays are produced in an x-ray. During a test run, the x-ray machine operates at a potential difference of 50kv. Each electron starts from rest and is accelerated towards the anode.

Meanwhile, a nearby industrial company uses a similar x-ray setup to inspect metal pipes for hidden cracks. However, both the hospital staff and industrial workers are warned about prolonged exposure to x-rays and protective measures are taken.

Additional data:

- Charge of an electron, $\boldsymbol{e} = 1.6 \times 10^{-19} C$
- Mass of an electron, $\boldsymbol{m} = 9.11 \times 10^{-31} kg$

Task

a. Justify how x-rays are produced in an x-ray and how they are effective in medical diagnosis and explain how the acceleration of electrons through given potential difference determines their kinetic energy.
b. Account for the associated health hazards and the need for safety precautions.

SECTION B

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ITEM 2

A telecommunication company in Kabale is planning to launch a network of satellites to improve internet and broadcasting services across East Africa. The project includes a main satellite that must remain fixed above one point on Earth and several smaller relay satellites that will transmit signals between remote areas.

During the planning meeting, one of the engineers compares Earth's satellites to the natural satellite, Moon. He explains that unlike Earth, the Moon has no significant atmosphere, which affects how signals and radiation behave around it. This raises important considerations for satellite placement and communication efficiency.

They also note that signal transmission between satellites involves electromagnetic waves travelling at the speed of light, and large orbital distances can introduce noticeable delays.

Additional data:

- Mass of Earth, $M = 6.0 \times 10^{24} kg$
- Radius of Earth,$R = 6.4 \times 10^{6} m$
- Gravitational constant,$G = 6.67 \times \frac{10^{-11} Nm^2}{Kg^2}$
- Time period,$T = 24 hours$

Tasks

a. Explain how signals are transmitted from satellites to earth. Justify how the distance of the main satellite affects the signal transmission between multiple satellites.
b. As a physics student, why the moon has no significant atmosphere for signal transmissions?

ITEM 3

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In Wakiso district, a team of engineers is designing a new water distribution system for a multi-storey hospital. The system includes storage tanks placed at a certain height and pipes of varying cross-sectional areas to supply water efficiently to different wards, including surgical units where steady pressure is critical.

During testing, the engineers observe that water flows faster through narrower sections of the pipe. They also need to ensure that the pipes can withstand internal pressure without deforming, which requires understanding the mechanical properties of the material.

One section of the pipe has a cross-sectional area of $4.0 \times 10^{-4} m^2$ and water flows with a speed of 2 m/s. at a narrower section, the area reduces to $2.0 \times 10^{-4} m^2$.the pipe material has a Youngs modulus of $2.0 \times 10^{11} Pa$ and engineers must ensure that any stress caused by pressure difference does not exceed the elastic limit.

Additional Data:

- Density of water,$\rho = 1000 kgm^{-3}$
- Acceleration due to gravity, $g = 9.8 ms^{-2}$

Tasks

a. Justify how the mechanical properties of the pipe material ensure water flow in the system by determining the velocity and pressure changes.
b. How do this pressure variations relate to stress, strain, and elasticity of the material in practical applications.

SECTION C

ITEM 4

At the Mulago health center, engineers are testing a neonatal incubator designed to maintain a constant temperature for premature babies. The incubator is enclosed with glass walls, and an internal heater provides warmth through both conduction and radiation.

Heat is lost through the glass walls by conduction, while the heater emits thermal radiation toward the baby and the surroundings. the engineers must ensure that the heat supplied balances the heat lost to maintain a stable temperature.

Given Data:

- Area of glass wall,$A = 0.5\ m^2$
- Thickness of glass,$d = 0.01\ m$
- Thermal conductivity of glass,$k = 0.8\ Wm^{-1}K^{-1}$
- $T_1 = 37^0C$
- Outside temperature,$T_2 = 27^0C$

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- Heater surface area, $A_h = 0.2\ m^2$
- Heater temperature, $T_h = 310K$
- Surrounding temperature, $T_s = 300K$
- Stefan-Boltzmann constant, $\sigma = 5.67 \times 10^{-8} Wm^{-2}K^{-4}$

Tasks

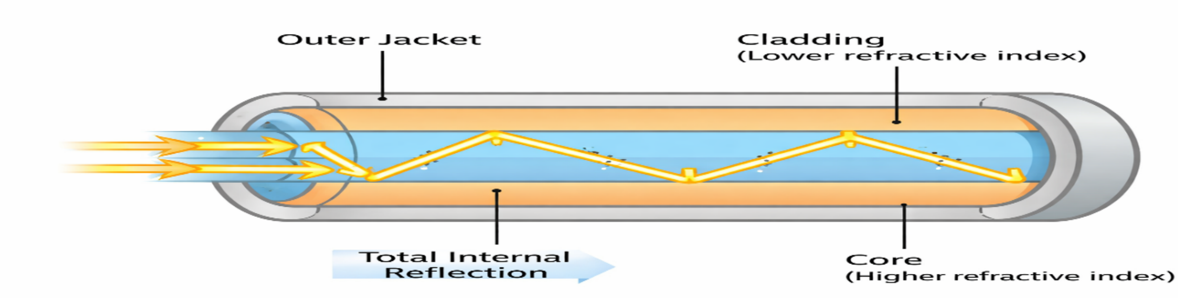
a. What is the ability of the incubator to maintain thermal stability by calculating the rate of heat loss through the glass and the net power of radiation emitted by the heater. Justify your response.
b. Explain how both thermal conductivity and radiation work together to regulate temperature in this medical application.

ITEM 5

At a science and technology center, students are demonstrating how sound and light waves are applied in modern communication systems. A stretched string fixed at both ends is made to vibrate using a mechanical oscillator, forming stationary waves. At the same time, an optical fiber system is used to transmit signals using light.

The instructor explains that the vibrating particles of the string execute Simple Harmonic Motion, and Stationary waves are formed due to the superposition of two waves travelling in opposite directions. The distance between two consecutive nodes is measured as 0.25 m.

In the same setup, light signals are transmitted through an optical fiber using Total internal Reflection. The core of the Fiber has a refractive index of 1.5, while the cladding has a refractive index of 1.4. for efficient transmission, the angle of incidence must exceed the critical angle.



Additional data: frequency of vibration of the string=200Hz

Tasks

a. Justify how wave propagation in the system supports efficient communication by calculating the wavelength, wave speed of the stationary wave.

b. Explain how the particles execute simple Harmonic Motion, and determine the critical angle for optical fiber and hence explain how Total Internal Reflection ensures effective signal transmission.

SECTION D

ITEM 6

In a busy phone charging center in Kampala, Musoke designed a high-efficiency charging station that allows many customers to charge their phones simultaneously. The system uses an arrangement of capacitors to stabilize voltage fluctuations caused by multiple chargers being connected and disconnected at different times.

Each charging unit includes a capacitor bank connected in parallel to store charge and smoothen the output voltage. The main power supply feeds the system through a potential divider circuit used to used to regulate voltage delivered to different charging ports.

During peak hours, up to 10 phones are connected, each drawing current from the system. The engineers must ensure that the capacitors can store enough charge to maintain steady output and that the system delivers maximum power safely without overheating.

Given Data:

- Supply Voltage,$V = 12V$
- Two resistors in potential divider$R_1 = 4k\Omega$, $R_2 = 8k\Omega$
- Total capacitance of capacitor bank, $C = 2000\mu F$
- Charging time 50minutes
- The total load resistance of connected phones $R_L = 6k\Omega$

Tasks

a. Justify how the capacitor arrangement improves stability in charging center
b. Explain how energy storage smooths voltage fluctuations and hence determine the power delivered to the load.
c. Explain the condition for maximum power and get the expression used and also showing how the system is optimized for heavy charging demand.

ITEM 7

A phone charging center in a busy trading area in Mubende is supplied with 240 V(rms), 50Hz AC from the national grid. To safely power multiple charging points, the center uses a step-down transformer, followed by a full-wave bridge rectifier and smoothing capacitor to convert the alternating current into a steady direct current suitable for charging phones.

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The transformer has a primary coil of 1200 turns and secondary coil of 100 turns. each phone charge requires a stable 5V DC supply and draws a current of 0.5 A during charging.

Tasks

a. As a physics student, justify:
b. The principle of operation of a transformer, and how the turns ratio affects the voltage transformation. Hence show how it suitable for stepping down voltage.
c. How the full-wave bridge rectifier converts AC to DC using diodes and how the workings for the peak secondary voltage and ideal DC output voltage.
d. Explain why the final output becomes suitable for charging phones.

END

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